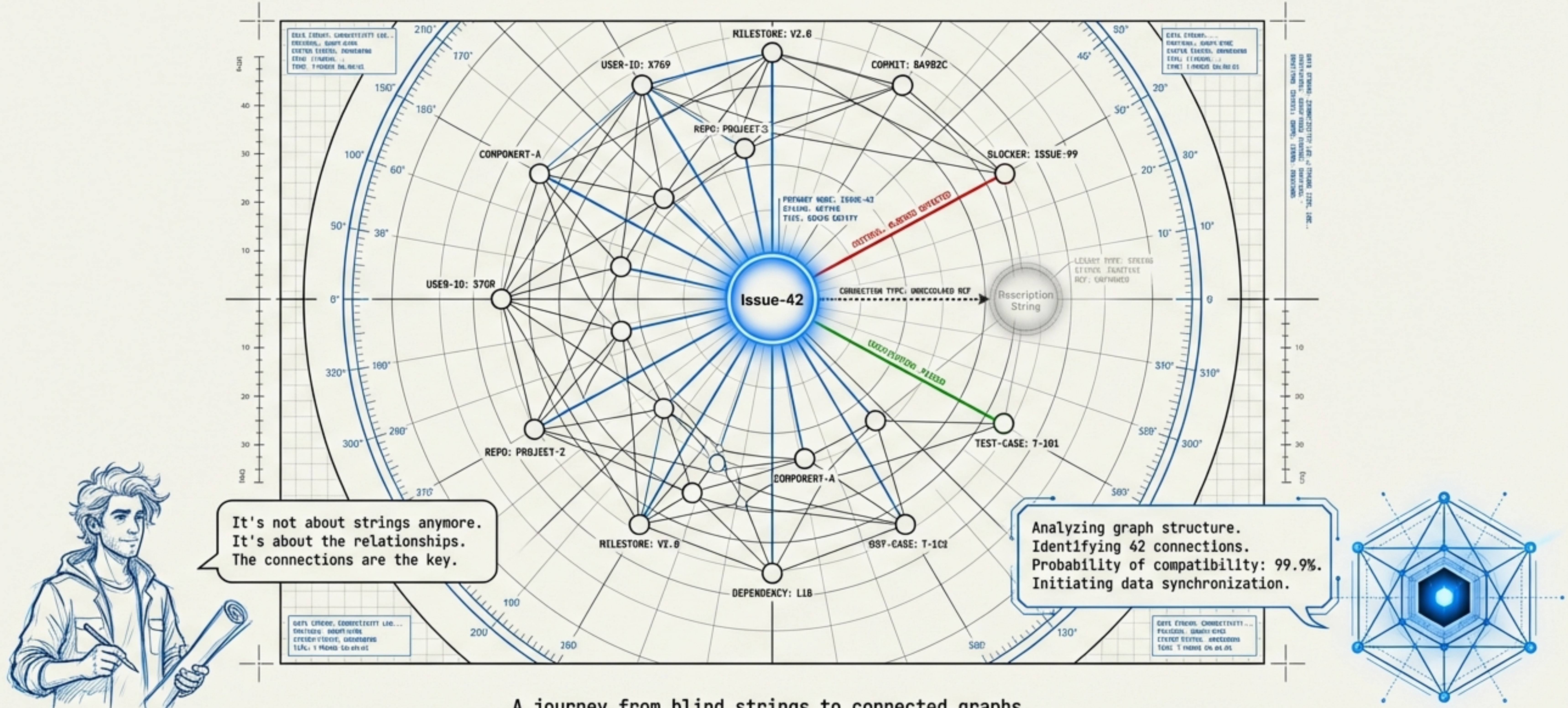
THE GRAPH AWAKENS

Compatibility Through Connectivity

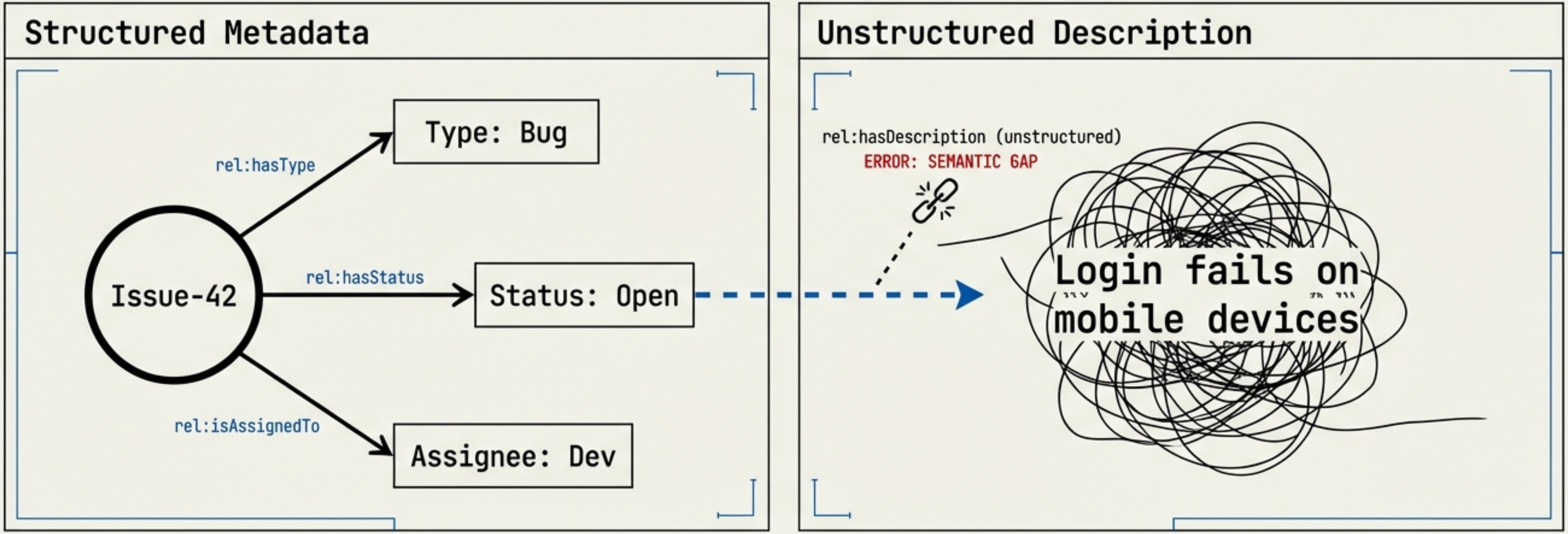


Dinis (The Architect)

A journey from blind strings to connected graphs.
Featuring Dinis (The Architect) and Claude (The AI).

Claude (The AI)

THE METADATA IS BEAUTIFUL, BUT THE GRAPH IS BLIND



Dinis

The edge connects to... nothing. Just characters. The graph can't see what they mean.

Unseen Edge

Error: Implicit Knowledge Loss

So a ‘Bug’ issue could describe a feature request, and the system would never know. The inconsistency hides in plain sight.

Claude

THE EXTRACTION DISASTERS

Without a map, extraction is just noise.

Attempt 1: LLM (No Guidance)

> Prompt: Extract everything.

the use's angand turkey
one is maybe land login
grapa
uselus login fails on
stane revord novr post next
tayaly tag not season
now sbul

Attempt 2: Standard NLP

> Login fails on mobile.

Entity:
Login ——> [PERSON]
Mobile ——> [ORGANIZATION]

WHO IS MOBILE AND WHY ARE THEY A COMPANY?

Attempt 3: Regex

```
> /^([a-z0-9_\-]+)@([\da-z\.-]+)\.([a-z\.]{2,6})$/
```

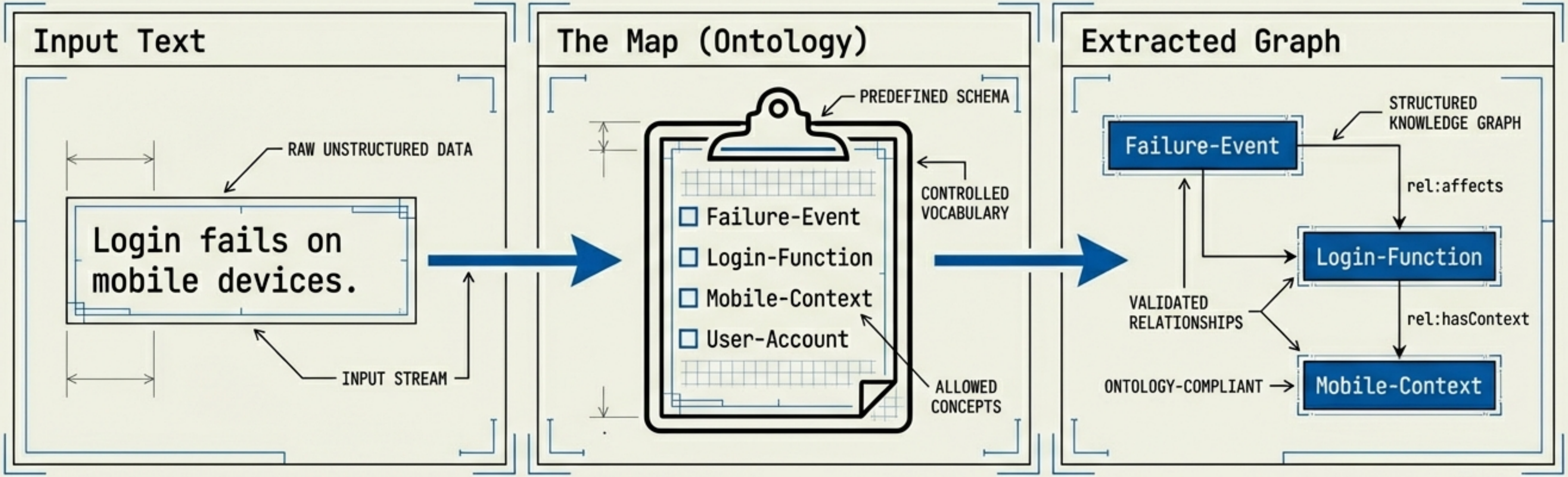
CATASTROPHIC BACKTRACKING ERROR

Dinis

Claude

THE ONTOLOGY IS THE CONTRACT

O&T First. Extraction Second.



It is like searching for treasure without a map. You must define what you expect to find. If a node exists in the graph, it must exist in the Ontology.

Claude

THE ‘GHERKIN PROBLEM’

Why text-based testing is brittle.

Scenario A

User sees the login page.

Match Found.

TEST PASS

Login ≠ Sign-in

Scenario B

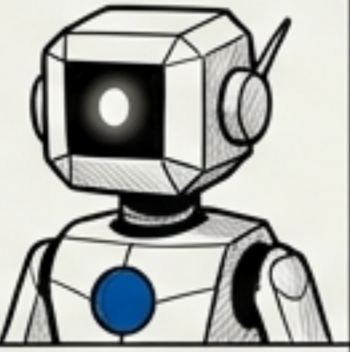
User sees the sign-in page.

Match Failed.

TEST FAIL

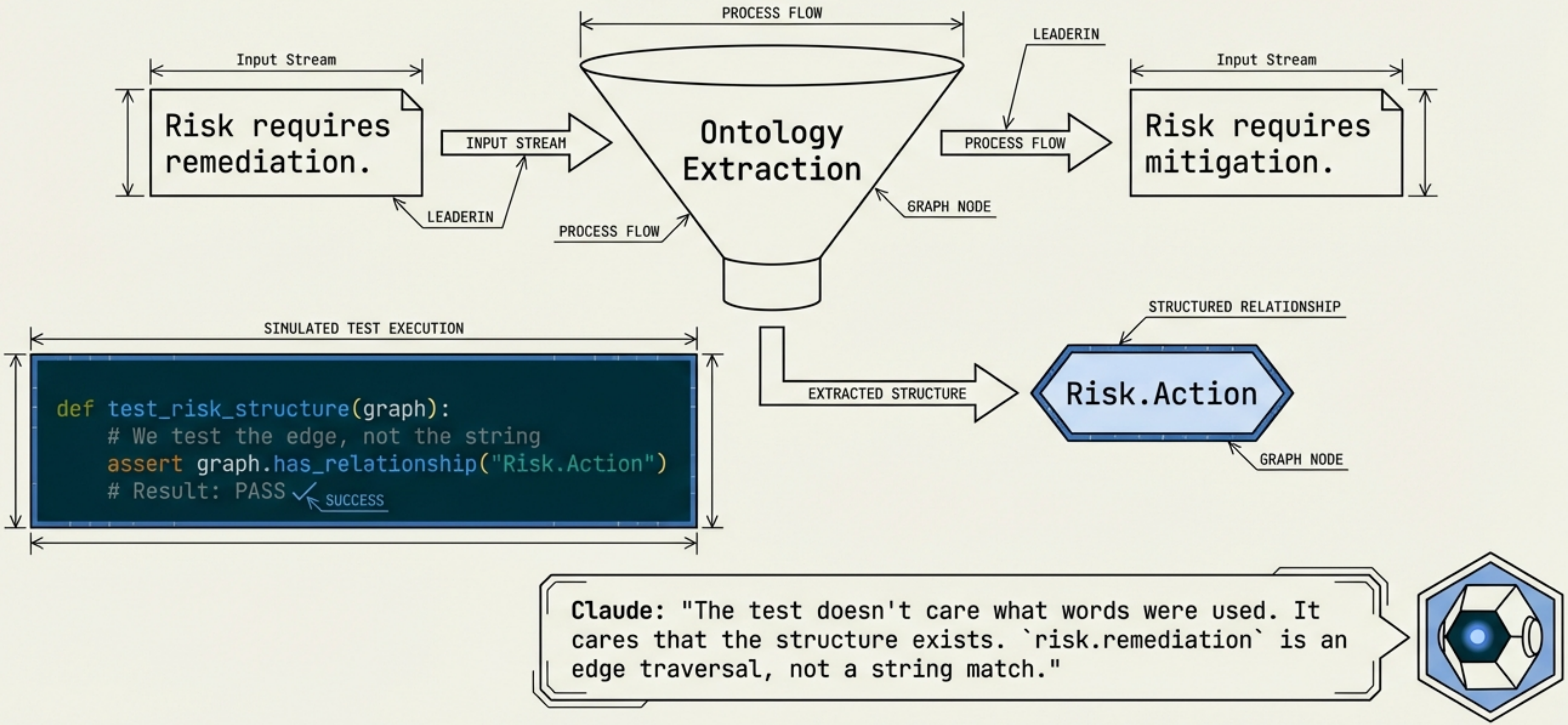
Computers don’t know synonyms.

Dinis
If we match words, we’re stuck. If the button text changes from ‘Submit’ to ‘Sign In’, everything is on fire.



WE DON'T TEST THE TEXT. WE TEST THE GRAPH.

Structure == Structure

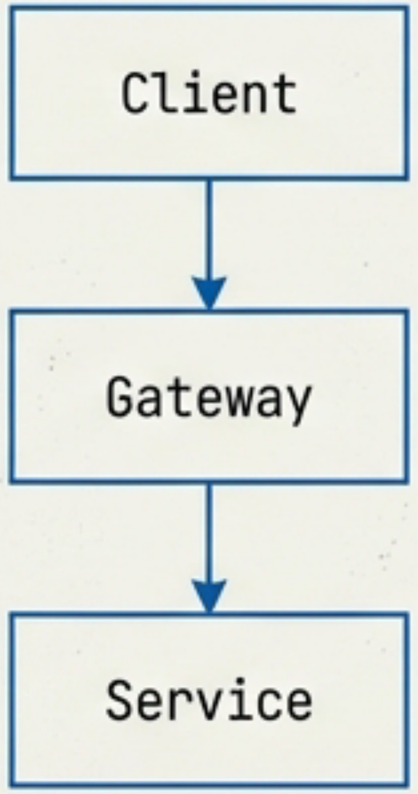


THE ARCHITECT'S LAMENT

The drift between fiction and reality.

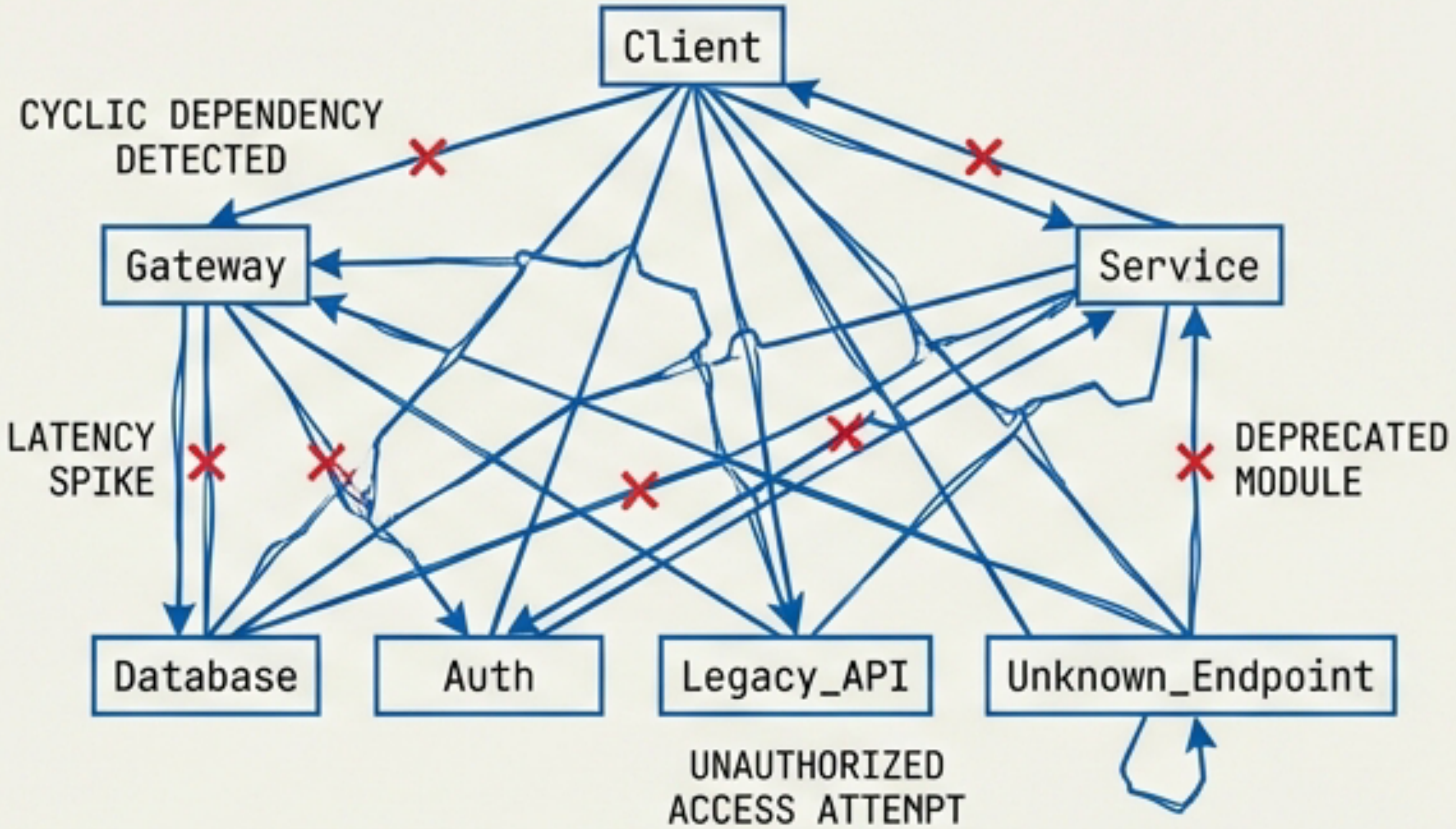
FICTION

Confluence Diagram (Last updated: 2 years ago)



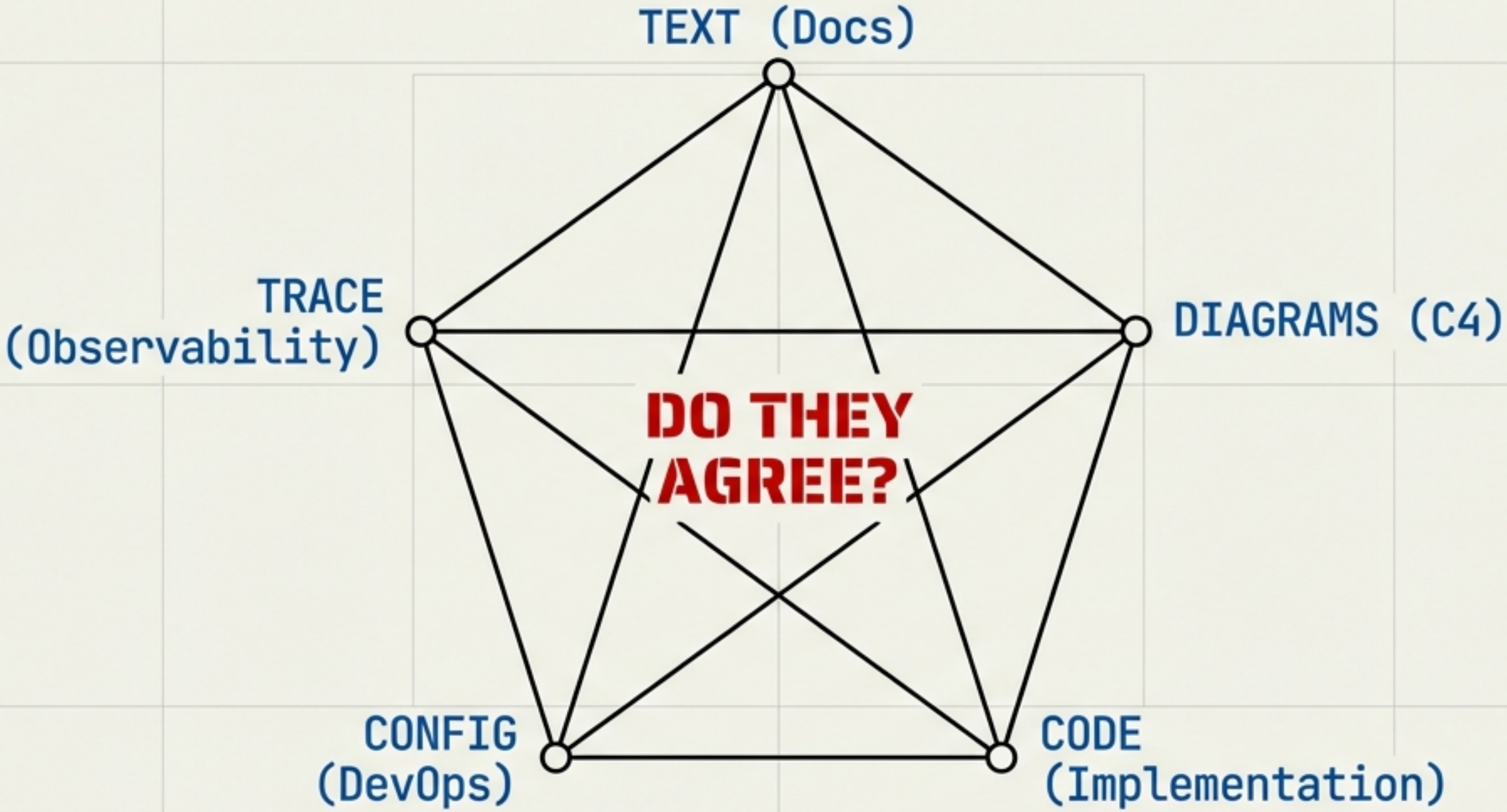
REALITY

Actual Codebase



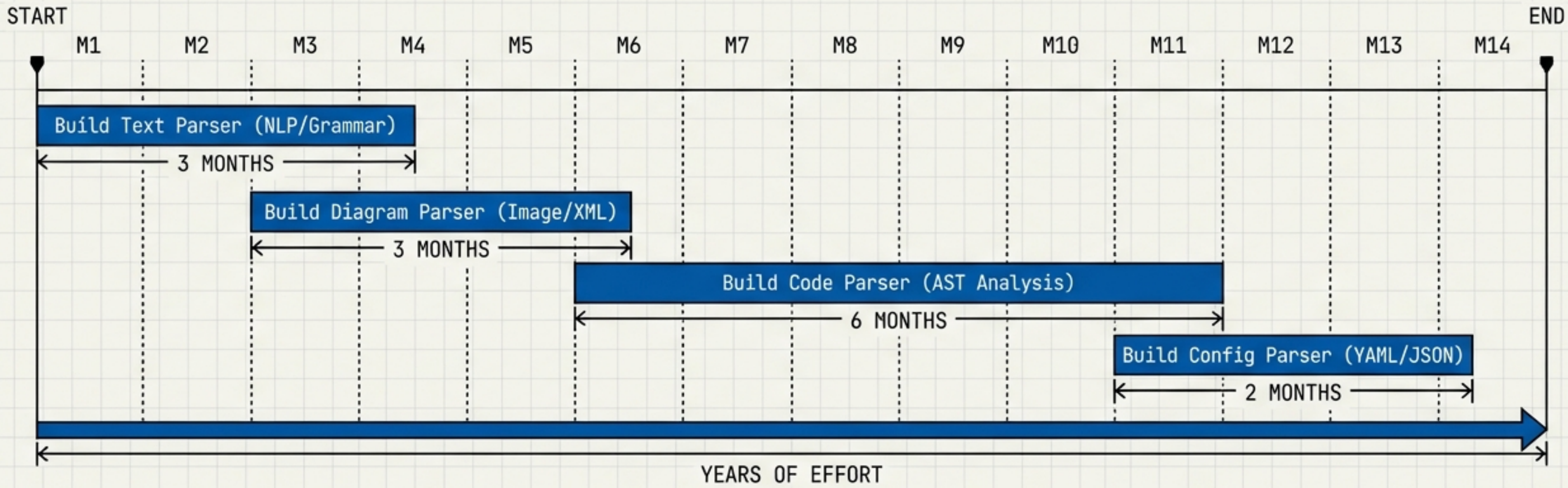
[redacted] I'm drawing fiction.
Beautiful, outdated fiction.

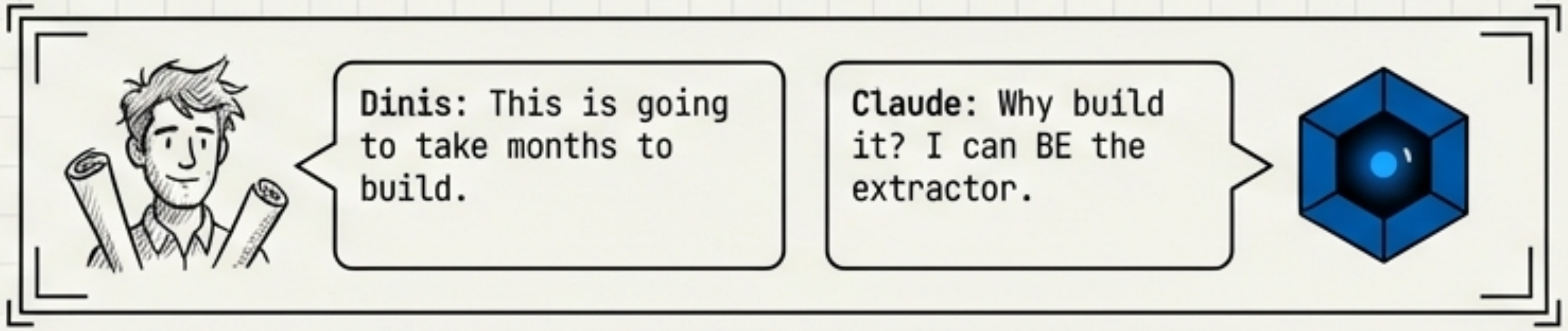
FIVE LANGUAGES DESCRIBING ONE SYSTEM



Claude: "They should all agree. But nobody checks. We must extract graphs from ALL of them and compare."

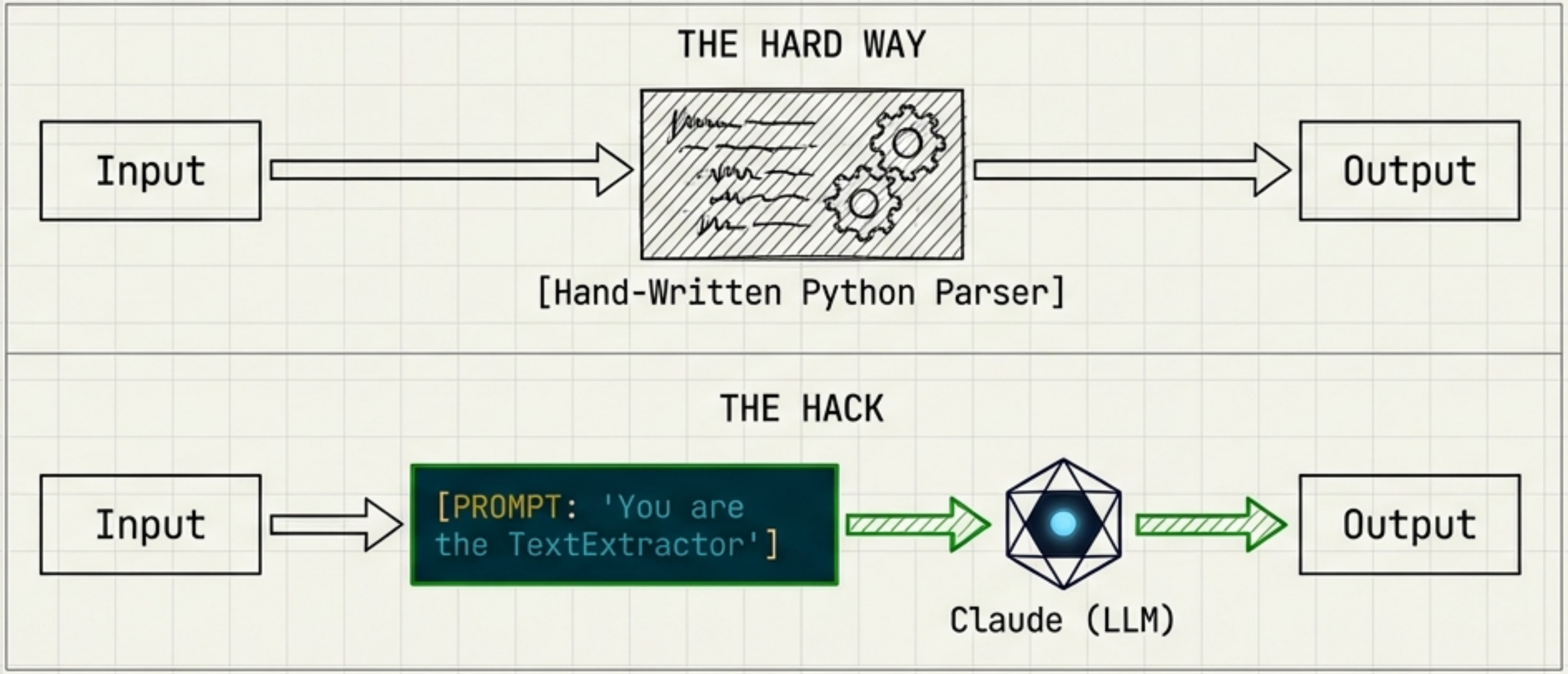
THE IMPLEMENTATION GAP





THE LLM AS EXECUTION ENGINE

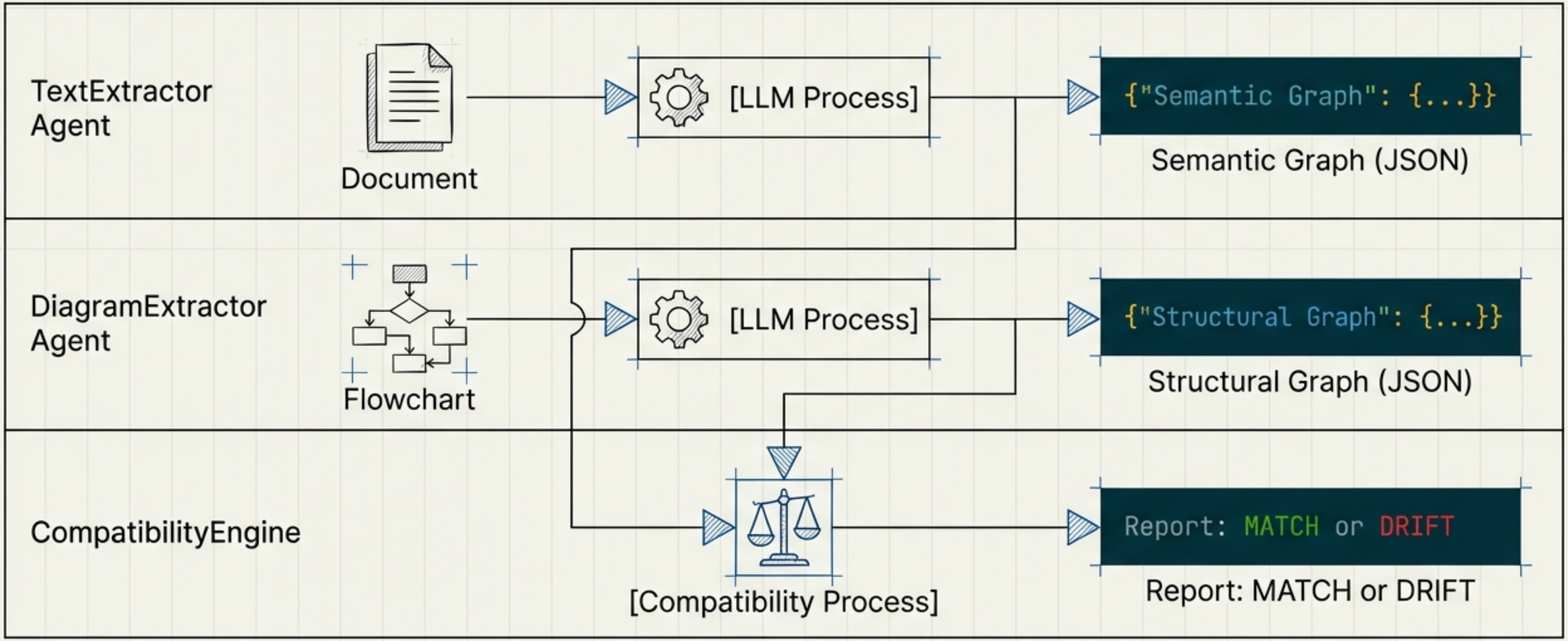
The Prompt is the Specification.



Claude: "I believe this is what you call a 'hack'. The prompt is the specification. I am the implementation."

THE AGENT WORKFLOW

From Artifact to Agreement



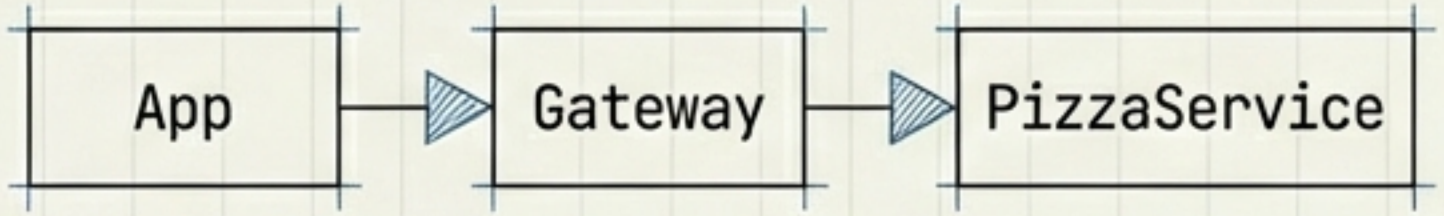
CASE STUDY: FOOD DELIVERY PLATFORM

When the layers agree.

The Rule

All external traffic must pass through the API Gateway.

The Diagram



The Extraction

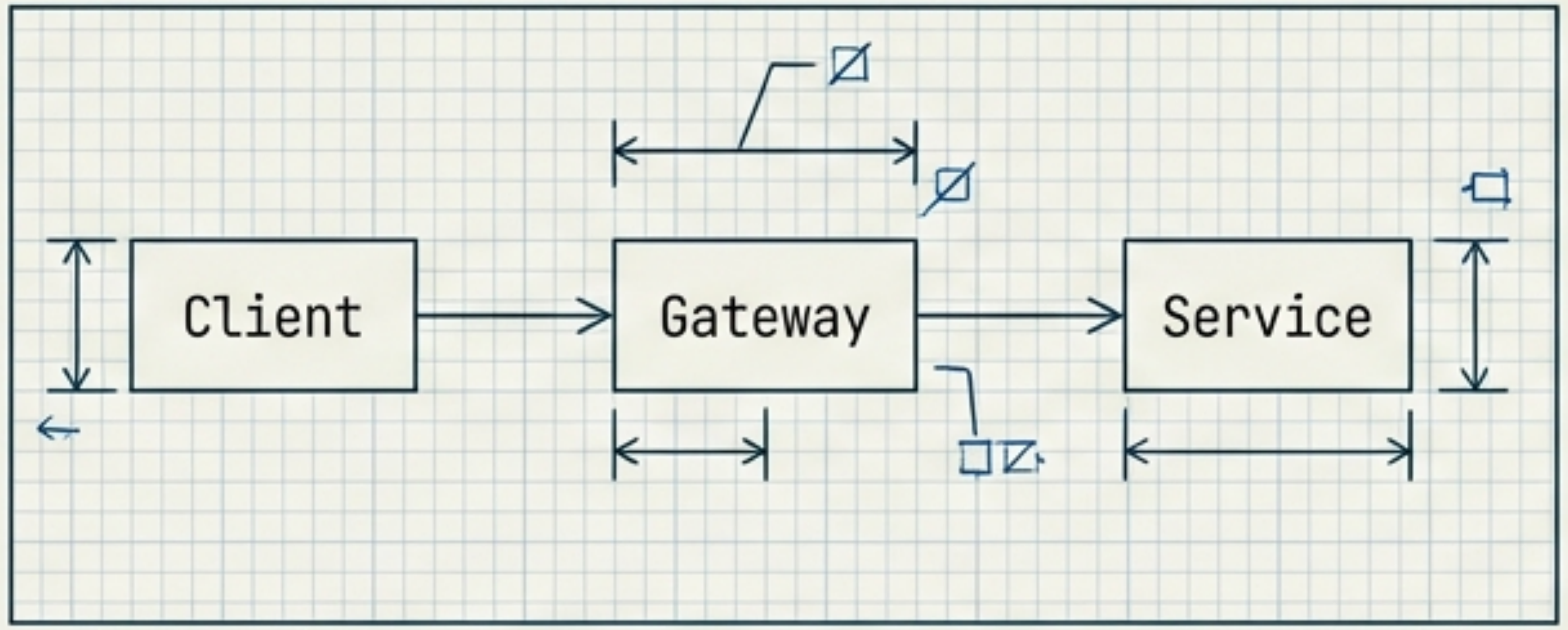

```json
{
  "source": "External",
  "target": "Gateway",
  "must_pass": true
}
```


AGREEMENT VERIFIED


```json
{
  "edge": "App->Gateway",
  "edge": "Gateway->PizzaService"
}
```

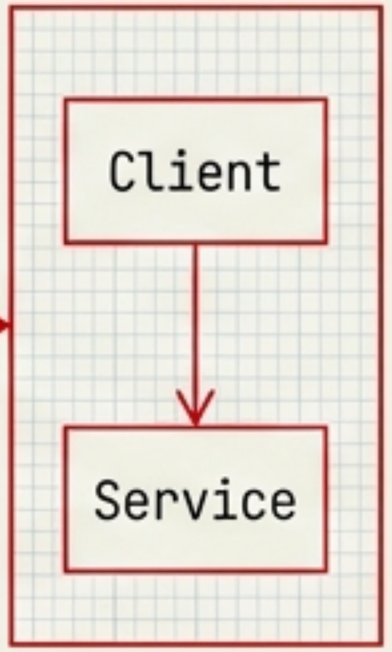
DETECTING THE DRIFT

The ‘Dave’ Scenario



INTENDED ARCHITECTURE

```python
# Dave's quick fix
def bypass_handler(request):
    return http.post("http://service-direct/api", data=request)
```

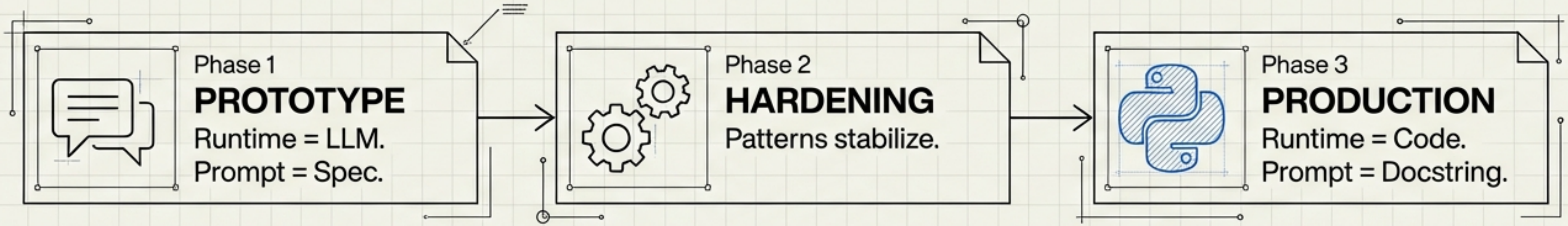


DISCREPANCY DETECTED. Code contains path 'Client -> Service' prohibited by Diagram.

Now we know. The diagram says one thing.
The code does another.

THE ‘CLAUDE LEGACY MODE’

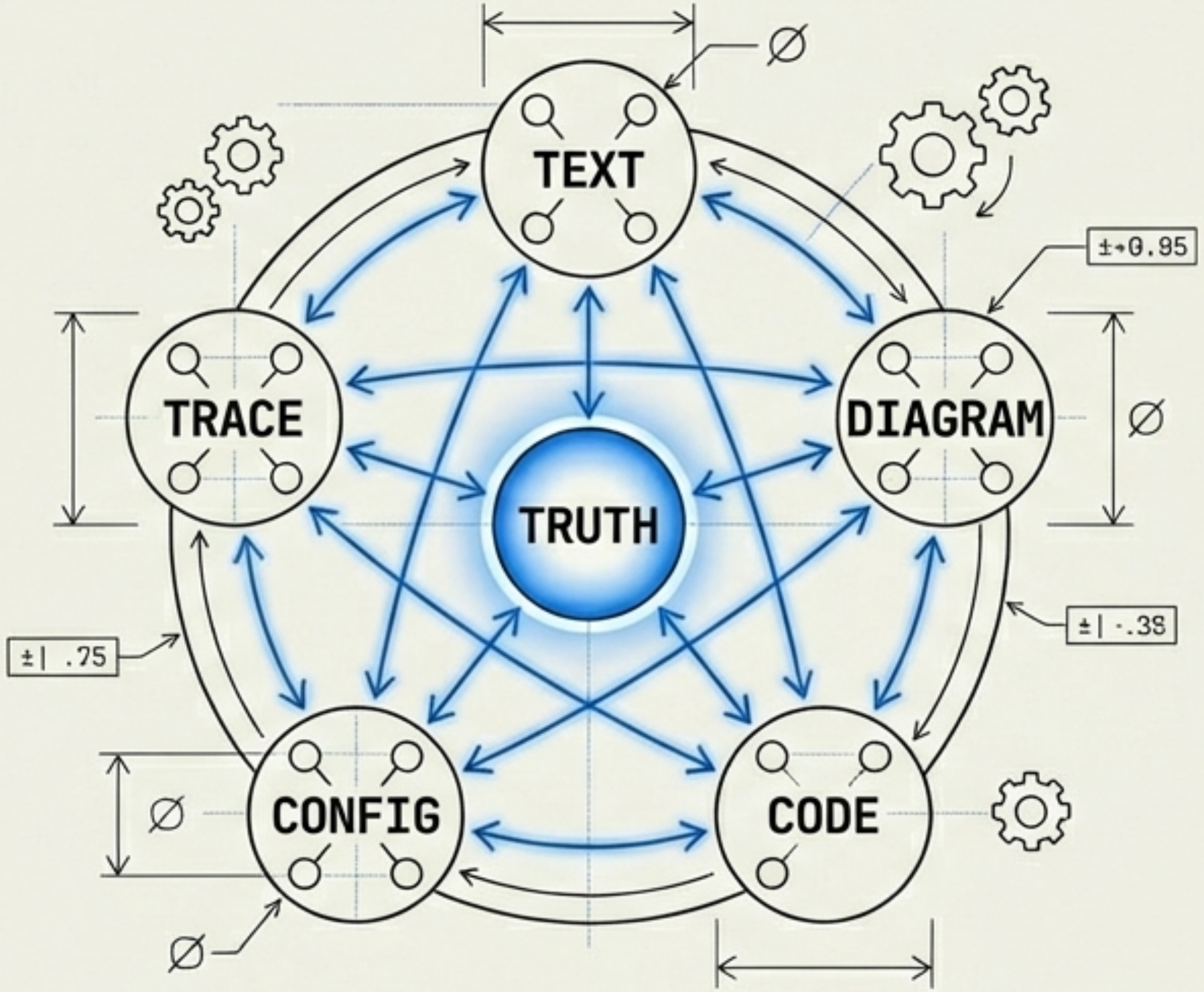
From Prompt to Python




```python
def text_extractor(text):
    """
    Original Prompt:  You are the TextExtractor...
    (Now implemented in pure Python)
    """
    return logic.parse(text)
```

I'm a version. There will be new versions. My role prompts will become Python functions. The circle of software life.

MEANING REQUIRES AGREEMENT



KEY TAKEAWAYS

1. Compatibility through Connectivity.
2. Do not assume; verify across layers.
3. Transform intellectual satisfaction into practical tools.
4. And Dave needs to read the docs.

We transform the intellectual satisfaction of solving a hard problem into a tool we can use today.